1(a) at t = 10.0 s,

its velocity towards East $v_E = 0.141 + (0.100)(10) = 1.14\ m\ s^{-1}$

its velocity towards North = 0.141 m s^{-1}

its resultant velocity $v_R = \sqrt{(1.14)^2 + (0.141)^2} = 1.15\ m\ s^{-1}$

at $\theta = \tan^{-1}(1.14/0.141) = 82.9^o$

(b) displacement towards East,

$x = 0.200 \cos 45^o (10.0) + \frac{1}{2}(0.100)(10.0)^2 = 6.41\ m$

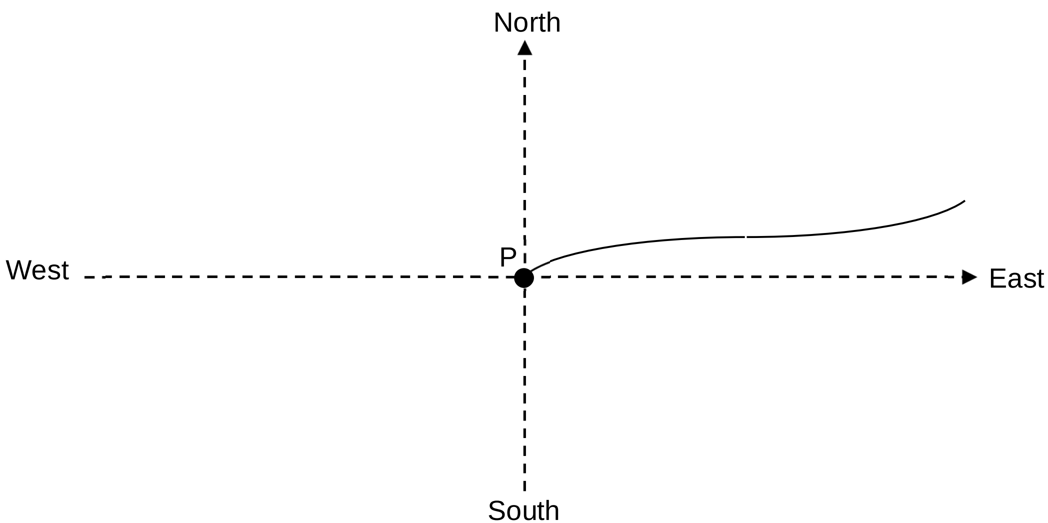
displacement towards North,

$y = 0.200 \sin 45^o (10.0) = 1.41 m$

Distance from point P , $r = \sqrt{(6.14)^2 + (1.41)^2} = 6.57 m$

(c) Straight-line with negative gradient. Non-zero ending.

(d)



2. **(a)** Taking the two particles as A and B, according to **Newton's 3rd law**, when